

Akshat Sureshbhai Desai

714-519-7380 | akshat.desai.754@gmail.com | linkedin.com/in/akshat-desai | github.com/Akshat1661 | Portfolio

EDUCATION

California State University, Fullerton

Master of Science in Computer Science

Fullerton, CA

Aug. 2024 – May 2026

EXPERIENCE

Lead AI Developer (Lightwall Project)

Oct 2025 – Jan 2026

Worthwhile Adventures LLC (Collaboration)

Fullerton, CA

- Built a **real-time multimodal ML system** integrating vision, audio, radar, and LLM-based control.
- Designed a **C++/Arduino inference bridge** for low-latency, hardware-efficient deployment.
- Applied **RAG grounding and prompt constraints** to reduce hallucinations and stabilize inference.
- Optimized inference paths with **sub-500 ms gating** and **2.8–4.1 s** end-to-end latency.

Graduate Research Assistant

Nov 2024 – Present

California State University, Fullerton

Fullerton, CA

- Processed **1,231 MRI volumes** on **HPC (SLURM)**, improving multi-class accuracy from **69.11% to 85.45%**.
- Built **RAG-based ML assistants** using ChromaDB and LLMs for code and design verification.
- Implemented automated evaluation and retry loops to enforce correctness in generated outputs.
- Designed a **LightGBM time-series forecasting pipeline** with feature engineering ($R^2 = 0.96$).

Machine Learning Engineer

Dec 2023 – May 2024

Space Applications Centre (SAC-ISRO)

Ahmedabad, India

- Designed **computer vision algorithms for satellite cameras** under dynamic lighting and distance constraints.
- Implemented **autofocus algorithms** using Laplacian sharpness metrics and focus-point optimization.
- Benchmarked Global, Binary, Hill-Climbing, and **Fibonacci** optimizers for real-time convergence.
- Developed **object-tracking-driven auto-exposure control** for real-time imaging systems.

PROJECTS

Histopathology Classification (Breast Cancer) | PyTorch, ResNet-50

Feb 2025 – May 2025

- Built a **PyTorch ResNet-50 pipeline** with transfer learning and inference optimization for multi-class classification.
- Achieved **99.86% AUC-ROC** and **98.61% specificity** across 8 visual classes.

Image Caption Generator | TensorFlow/Keras, VGG16, GRU, Dash

Oct 2023 – Dec 2023

- Built a **vision-language model** combining CNN encoders (VGG16) with GRU sequence decoders.
- Generated natural-language descriptions from images using end-to-end training and inference.

YOLOv8 Waste Detection System | YOLOv8, Computer Vision

March 2023 – May 2023

- Designed an **anchor-free real-time object detection system** using YOLOv8.
- Achieved **3–9 ms inference latency** using lightweight detection heads suitable for edge deployment.

AWARDS

1st Place — CSU Generative AI Hackathon (AWS)

Aug 2025

- Built a Flask & JS dashboard serving **XGBoost** staffing predictions via single-day and batch APIs.
- Integrated an **AWS Bedrock & LangChain** assistant for natural-language staffing insights.

TECHNICAL SKILLS

Languages: Python, C++, SQL

Machine Learning: Linear/Logistic Regression, SVMs, Transformers, XGBoost, LightGBM, Model Evaluation

Deep Learning / CV: PyTorch, TensorFlow/Keras, ResNet, YOLOv8, OpenCV, CUDA, Mixed Precision, Detection, Tracking, Segmentation

Generative & Multimodal Models: Vision-Language Models, CLIP-style Architectures, LLMs, Hugging Face, vLLM, RAG

Efficient Models / Systems: Quantization, Model Pruning (conceptual), ONNX Runtime, Inference Optimization, Linux, Docker, SLURM, AWS, Cloud Run